SP-VIVIFY-3: Implemented, Rejected as a Default

Tactix Development Project

9 August 2026

Executive summary

Tactix now has CaDiCaL's schedule-level Vivify prefix flush, with the proof-rule/strategy boundary preserved. The implementation is proof-clean and well exercised, but a same-window Tier C comparison loses two solves and 164.46 PAR-2; the sign survives removal of any four instances. The mechanism is therefore retained behind `TACTIX_VIVIFY_FLUSH=1` and ships off. This decision also avoids importing a CaDiCaL-specific default into the immediate Kissat-default campaign.

1 Implementation and boundary

Commit `a7d7a4b` orders Vivify candidates by occurrence-ranked literal sequences within four separate schedule bands. If the preceding retained sequence is a prefix, the larger clause is removed before propagation. The deletion is ordinary C1 subsumption and uses the shared proof deletion path.

The public N1 surface did not change: candidate enumeration and per-clause instances remain complete and side-effect free. Only the canonical round may flush, and only when the strategy channel enables it. Focused tests cover the enabled and disabled behaviors, proof deletion, and separation of irredundancy and tier bands.

2 Correctness evidence

The Release suite passes 60/60 after switching the default off. Earlier focused assertion and address/undefined-sanitizer tests passed 6/6. Final enabled-pin LRAT fuzz job 1594317 ran 10,000 cases with zero failures and reached the new operation in 1,416 cases. Targeted job 1594318 ran another 10,000 cases with zero failures, forced 2,784,200 Vivify applications, and reached the flush in 1,942 checked cases. Harness self-test 1594319 passed.

Every scored Tactix SAT model and UNSAT proof in the benchmark windows was checked. All three result files report clean correctness: no mismatch, invalid model, rejected certificate, checker give-up, or unverified scored result.

3 Performance evidence

The four-arm arrays compared enabled, disabled, Kissat, and CaDiCaL in the same window. They used commit `a7d7a4b`, binary SHA-256 `01647eba79bbb9cb...e6ec774998`, the standard partition, and no array throttle. Tier A/B used one core and 8 GB per task; Tier C used two cores and 16 GB.

Tier	Enabled	Disabled	Kissat	CaDiCaL
A	200/200, 0.52	199/200, 0.67	200/200, 0.45	199/200, 0.75
B	28/28, 52.48	26/28, 69.52	28/28, 10.06	27/28, 40.54
C	19/30, 1047.71	21/30, 883.25	29/30, 249.42	28/30, 379.15

Each cell is solved count then PAR-2. On commonly solved cases, enabled/disabled geometric-mean runtime ratios are 0.839 (A), 1.214 (B), and 1.186 (C). Tier C's disabled arm additionally solves `StConn_8_32` and `j3037_9_rggt_b`; enabled times out on both. Five instances must be removed to reverse the Tier C sign. Reach counters show that the feature is not inert: it removes 107,779, 1,115,949, and 2,061,500 candidates on A, B, and C.

Primary evidence is `benchmarks/reports/2026-08-09-sp-vivify-3.md`; it records the three raw run directories, immutable hashes, and per-instance transitions.

4 Decision, risks, and next action

The pre-registered rule said that a broad Tier C loss blocks default adoption. That condition fired, so prefix flushing ships off. The retained on arm is useful for future interaction studies, but the current data do not support a more elaborate admission policy. The result is specific to Tactix's current Vivify implementation and schedule; it is not evidence that the C1 deletion is unsound or that CaDiCaL's option is generally poor.

The next ledger item is `SP-SWEEP-3`: persist Sweep's ranked untried tail across calls and grow environment caps only after a completed pass. It should again be kept out of the proof-rule surface and priced with the smallest gate that can decide its default.